

Document Upload User Guide For NIH IC Partner Systems

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Purpose

The purpose of this document is to provide a general description and usage guide for eRA's Document Upload Web Service, used by NIH IC Partner Systems.

General Description

The Document Upload REST Web Service is for systems to upload documents into eRA database using HTTP protocol.

security

Secure Socket Layer (SSL) client certificate is required for authentication and authorization. The ACCESS_DOC_UPLOAD_SERV role is needed to upload documents.

Pre-requisites

The caller must have a certificate from an authorized certificate provider that has been configured/registered for accessing this service. Information on the usage of certificates can be found at https://grants.nih.gov/grants/electronicreceipt/files/S2S_Web_Services_Guide.pdf

The certificate used to connect to the service must be mapped to an IMPACII account that has ACCESS_DOC_UPLOAD_SERV role via User Admin as part of the account registration process.

Service locations

- Test-- <https://services.internal.test.era.nih.gov/docservice/dataservices/document>
- Staging – <https://services.internal.stage.era.nih.gov/docservice/dataservices/document>
- Production – <https://services.internal.era.nih.gov/docservice/dataservices/document>

Operations

Document Upload Web Service supports the following operations that conform to the HTTP 1.1 standard:

Operation	Input parameters	Output parameters	Role needed
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POST (Upload document)	docType id document file	HTTP Status code Message Token Response Time	ACCESS_DOC_UPLOAD_SERV
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Supported document types

The supported document types and ids are listed below:

docType	Description	Id	File MIMETYPE
CRU	Council of Councils Report	Appl_id	application/pdf
CCR	Council of Councils Recommendation	Appl_id	application/pdf
NPC	NCI Program Checklist	Appl_id	application/pdf
NGC	NCI Grants Management Checklist	Appl_id	application/pdf

Input Values and Output Values

POST

Description

The POST operation adds a new document to the eRA database. If the operation is successful, a token will be returned with a message. The token can be used for retrieving the audit information and other troubleshooting.

Requests

POST [/docservice/document/<docType>/<id>](#) HTTP/1.1Host: [services.internal.era.nih.gov](#)

User-Agent: *browser_data*

Accept: *file_types*

Accept-Language: *Regions*

Accept-Encoding: *encoding*

Accept-Charset: *character_set*

Keep-Alive: 300

Connection: keep-alive

Content-Type: **multipart/form-data; boundary=-----128861817228863**

Content-Length: 3278385

-----128861817228863

Content-Disposition: form-data; name="docFile"; filename="1.pdf"

Content-Type: application/pdf

%PDF-1.4

%âãïÓ

2 0 obj

<</Length 1310/Filter/FlateDecode>>stream

....

-----128861817228863

Request Parameters

The request parameters are in the URL and request body. The URL is like the following:

<https://services.internal.era.nih.gov/docservice/document/<docType>/<id>>

The parameters are listed below:

Name	Description	Parameter location	Required
<i>docType</i>	A 3 character string to indicate the type of the document. Type: String	URL	Yes
<i>Id</i>	A unique string to identify the document. Type: String	URL	Yes

<i>docFile</i>	Document file content. Type: multipart/form-data as specified in RFC 2388 (http://tools.ietf.org/html/rfc2388) and the following content header need to be specified: Content-Disposition: form-data; name="docFile"; filename="<submitted file name>" Content-Type: application/pdf The <submitted file name> will be used when the document is retrieved. The multipart name must be "docFile", and if there are multiple parts with the same name, only the first one will be taken.	Body, form field.	Yes
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Responses

In addition to the standard HTTP response codes, this operation returns the following response headers.

Name	Description
<i>x-docservice-token</i>	A token as confirmation of the operation. Type: String
<i>x-docservice-response-time</i>	The milliseconds it takes to process the request. Type: number

Response code and messages

Document Upload Web Service follows HTTP 1.1 standard. The following are the response codes, messages with their meanings.

HTTP status code	Message	Meaning
200	Document was uploaded successfully:<token>	POST success.
403	Access denied.	Insufficient privilege.
400	Invalid parameters.	Invalid parameters such as doc type, id, missing content.
415	File type not allowed.	File type not allowed for upload.
409	Document exists	POST cannot be done because of existing document.
500	Internal error.	Internal server error.

Sample HTTP trace

POST

Sample request:

```
POST /docservice/document/CRU/7878950 HTTP/1.1Host: services.internal.era.nih.gov
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; rv:30.0) Gecko/20100101 Firefox/30.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,/;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-----128861817228863
Content-Length: 3278385
-----128861817228863
Content-Disposition: form-data; name="docFile"; filename="1.pdf"
Content-Type: application/pdf
%PDF-1.4
%âãÏÓ
2 0 obj
<</Length 1310/Filter/FlateDecode>>stream
....
-----128861817228863
```

Sample response:

```
HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
Date: Tue, 01 Jul 2014 19:04:15 GMT
Content-Type: text/plain
x-docservice-token: ETLbewAAUFu
/bWBABRBRVMvQ0JDL1BLQ1M1UGFkZGluZWCAABAAEHOnPtJhSlqBApsuTeNDknoAAABWq4G8uL5pLoIrm8EYQYwn03Zgwo2PooVGh6svgQRBvpB
sEDiRYFlyV0PppquvJgRjS6d4UIQFREbQXdOQBGxPmPgLO+jyh55eE6hgT2we7VT+ukHWvIDEelrSynfrdikyTqStkMuhpQ/CKOwrJhgbLQAU5s0hYb6O5
/8o7sc15WnsqjVWkbQ=
x-docservice-response-time:555
Document was uploaded successfully.
```

Sample client

Upload through browser using a HTML page:

```
<html>
<body>
<form method="post" name="f" enctype="multipart/form-data" action="https://services.internal.test.era.nih.gov/docservice/dataservices/document/CRU
/7878950">
<p>
label to display:
<p/>
<p>
Please select a PDF (.pdf) file to be sent:
<p>
<input type="file" name="docFile" size="40" accept="application/pdf" />
<p>
<input type="submit" value="Send">
</form>
</body>
</html>
```

Internet Standards

The following are the standards about file upload through HTTP RESTful service.

Term	Standards	URL
HTTP Request Methods	IETF RFC 2616: Hypertext Transfer Protocol – HTTP/1.1	http://en.wikipedia.org/wiki/HTTP#Request_methods http://tools.ietf.org/html/rfc2616
Multipart Form Data	IETF RFC 2388: Returning Values from Forms: multipart/form-data	http://tools.ietf.org/html/rfc2388
HTML File Upload	IETF RFC 1867: Form-based File Upload in HTML	http://tools.ietf.org/html/rfc1867